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https://github.com/julianmak/OCES4303_ML_ocean

The repository principally contains the compiled products rather than the source for size reasons.

- ▶ Associated Python code (as Jupyter notebooks mostly) will be held on the same repository. The source data however might be big, so I am going to be naughty and possibly just refer you to where you might get the data if that is the case (e.g. JRA-55 data). I know I should make properly reproducible binders etc., but I didn't...
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OCES 4303 :
an introduction to **data-driven and ML methods** in ocean sciences

Session 8: intro to neural networks

Outline

- ▶ anatomy of a **neural network**
 - nodes, weights, hidden layers, activation functions
 - back-propagation
- ▶ **(multi-layer) perceptrons** (MLPs)
 - simple perceptrons and interface with linear models
 - solvers, loss curves, various options
- ▶ cats and dogs example: classification and regression

Oceanic application

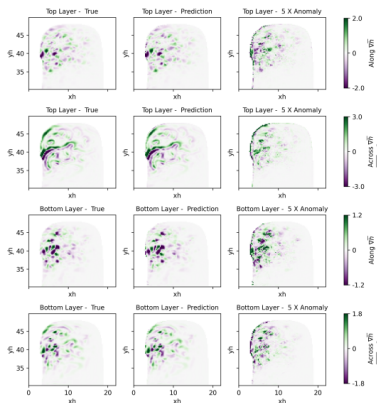


Figure: Model truth, predicted and mismatches in the predicted thickness fluxes from a double gyre calculation. From Balwada *et al.* (submitted).

- ▶ many instances where **neural networks** are used
→ but usually not in the MLP form for reasons to be elaborated
- ▶ example in predicting a spatially varying field, application in parameterisations
→ prediction here for **thickness** fluxes; see next sessions with **CNNs** for more

Recap: classifiers/regressors

- recall for supervised learning we basically have

$$Y = f(X),$$

and given X and Y we want the classifier/regressor f

- f can be represented
 - symbolically (e.g. linear models)
 - as decision trees (e.g. random forests)
 - as a chain of elementary operations (e.g. **neural networks**)

Neural networks: anatomy

- ▶ easiest probably with an example:

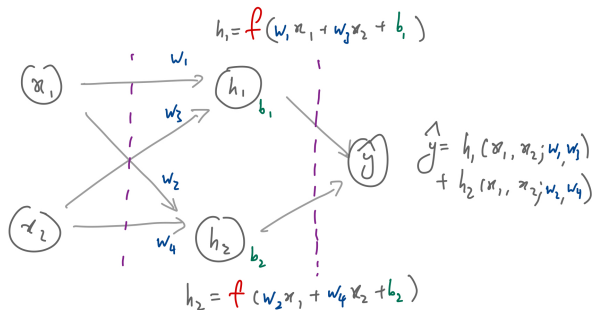


Figure: Made up example of a neural network (that you can code/train up yourself in principle; see notebook).

- ▶ input $X = (x_1, x_2)$ passes through network and gets transformed into prediction $Y = \hat{Y}$

Neural networks: anatomy

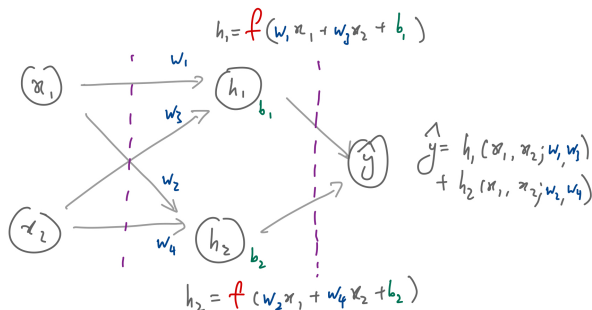


Figure: Made up example of a neural network (that you can code/train up yourself in principle; see notebook).

- each **hidden layer** have **nodes**, where we
 - multiply things by **weights** w_i
 - add on **biases** b_i
 - passed through an **activation function** f

Neural networks: anatomy

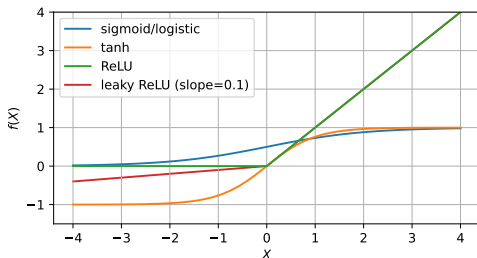


Figure: Samples of common activation functions: sigmoid (bounded between $[0, 1]$), tanh (bounded between $[-1, 1]$) and ReLU (bounded below by 0). Note ReLU is basically hinge loss but flipped the other way around. Leaky ReLU has a slope in the negative part.

- **activation functions** add nonlinearity to the system (see later)
 - maps the elementary operations to known ranges
 - usually piecewise differentiable (see three slides later for reason)
 - e.g. ReLU controls 'firing' of neurons (if $x < 0$)

Neural networks: as approximators

- Q. just a bunch of elementary operations, cannot be complex enough surely?
- ▶ **universal approximation theorem** for neural networks:
 - ‘sensible’ (!?) functions can be approximated to using above operations given sufficient ‘complexity’
 - ‘complexity’ means number of nodes and/or layers
 - ▶ note: an **existence** theorem
 - doesn’t tell you how you would train it
 - there are lower/upper bounds for complexity in some cases

Neural networks: training

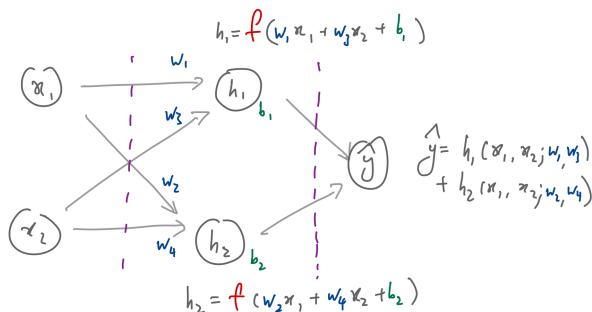


Figure: Made up example of a neural network (that you can code/train up yourself in principle; see notebook).

- one **feed-forward** results in $\hat{y}$
 - evaluate a **loss function** $J(Y, \hat{y})$
 - J depends on problem, e.g. one sided for classification, L^2 for regression, information entropy based etc.

Neural networks: training

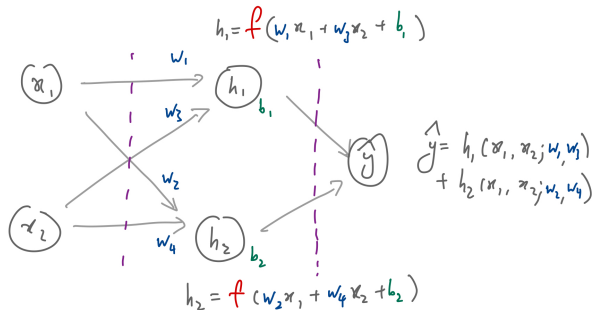


Figure: Made up example of a neural network (that you can code/train up yourself in principle; see notebook).

- aim: find minimum J by changing model parameters

$$\theta = (w_i, b_i)$$

→ want to solve something like $\partial J / \partial \theta = 0$ (abusing notation here...)

Neural networks: training

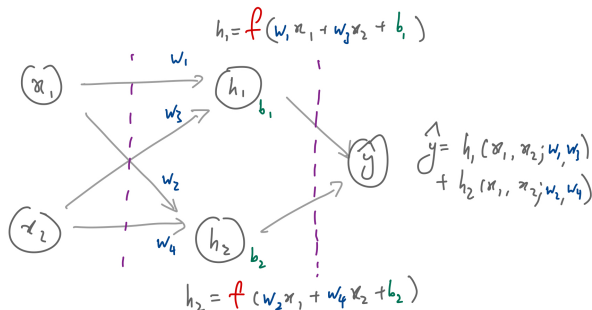


Figure: Made up example of a neural network (that you can code/train up yourself in principle; see notebook).

► **back-propagation** (chain rule basically): e.g.,

$$\frac{\partial J}{\partial w_1} = \frac{\partial J}{\partial h_1} \frac{\partial h_1}{\partial w_1} = \frac{\partial J}{\partial h_1} (f'(w_1 x_1 + \dots) + x_1 f(w_1 x_1 + \dots)).$$

→ do for all, SGD etc. and iterate until convergence/bored

Neural networks: training

- ▶ more nodes?
 - more equations
- ▶ more layers?
 - longer chains
- ▶ activation functions?
 - choose piecewise differentiable and 'easy' functions
- ▶ other components within the hidden layer?
 - (e.g. convolutional kernels in CNNS, hidden states in RNNs)
 - assume differentiability but proceed as usual
- ▶ constraint and/or dynamical layers?
 - can deal with in principle (e.g. adjoints), but beyond this course...

Perceptrons

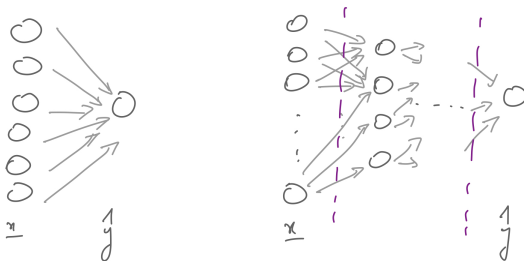


Figure: Schematic of perceptrons (no hidden layers) and MLPs (at least one hidden layer).

► perceptrons

- no hidden layers, may have activation functions
- can only really do binary classification (i.e. 'percept')
- case of no activation function $\Rightarrow$ matrix inversion essentially

Perceptrons: classification

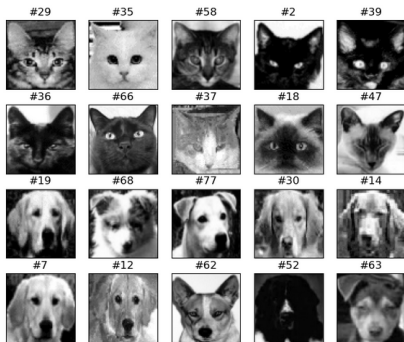
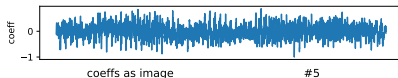
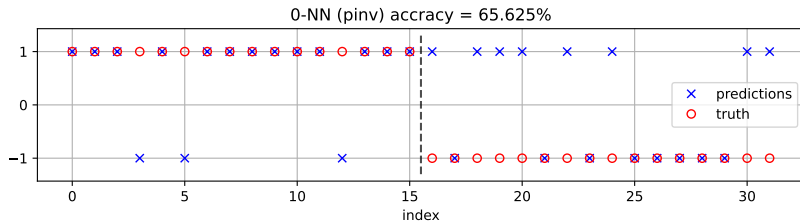


Figure: Predicted log of diatom-dinoflagellate ratio in HK coastal waters using random forests (which is a collection of decision trees) under projected warming scenarios. From Fig. 4 of Cheung *et al.* (2021).

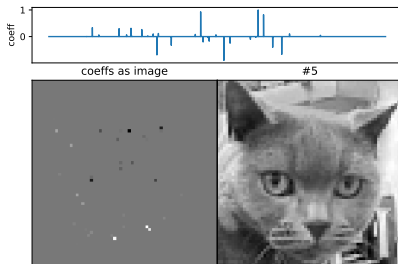
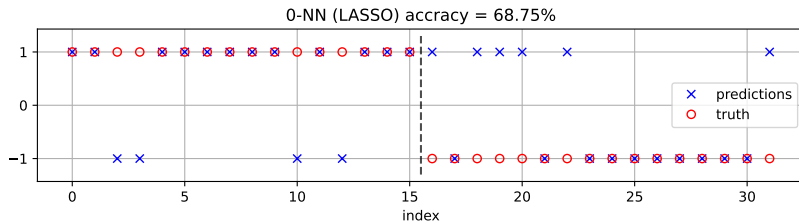
- classifying cats and dogs
 - this is a hard problem!
 - cats labelled as 1 and dogs as -1
 - throw in full image
 - standardise, train/test split
 - can probe the model in this case, **feature identification**

Perceptrons: classification



- ▶ just matrix inversion
 - L^2 minimisation with no penalisation
 - many non-zero values
 - some sort of 'shape', 'eyes' and 'facial features' identified?

Perceptrons: classification



- Lasso optimisation instead
 - L^2 minimisation with L^1 penalisation
 - promotes sparsity (but α dependent)
 - 'eyes', 'forehead' and 'mouth' important?

MLPs

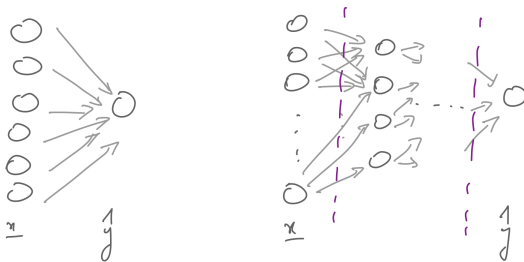


Figure: Schematic of perceptrons (no hidden layers) and MLPs (at least one hidden layer).

► multi-layer perceptrons (MLPs)

- bit of a misnomer, because above limitation with perceptrons do not strictly apply
- sometimes artificial neural networks (ANNs) (misleading?)
- can be 'deep' neural networks (DNNs) (misleading?)

MLPs: classification

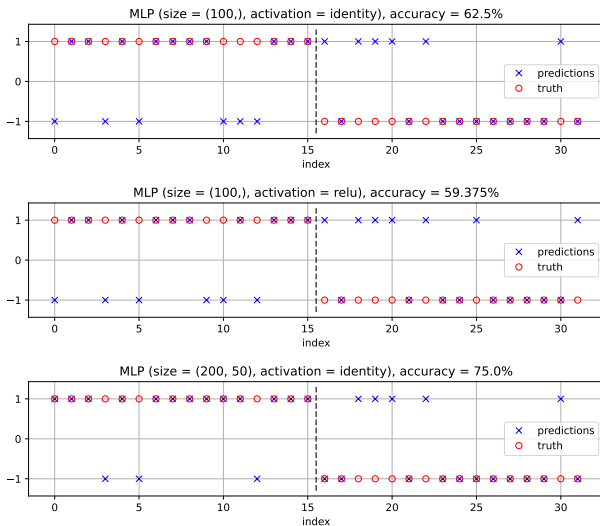


Figure: Three different attempts at using MLP to do full image classification.

MLPs: classification

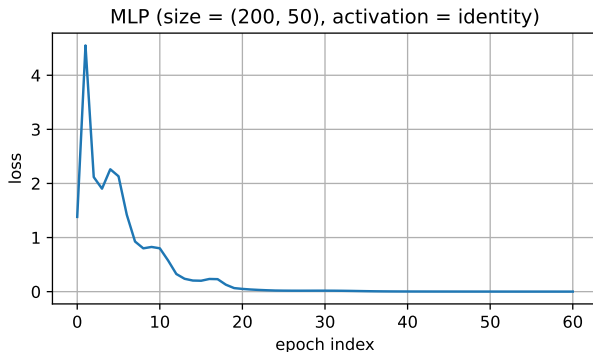


Figure: example of a loss curve with training epoch.

- ▶ an example of a **loss curve** with training decreasing with iterations (**epoch**)
 - continues until some criterion is reached
 - **early-stopping** is possible (but not considered here)

MLPs: regression problem

- ▶ just using cats
 - predict top half from bottom half
 - vice-versa might work better?
- ▶ again, this is a hard problem!



Figure: No animals were hurt in the present demonstration.

MLPs: regression problem

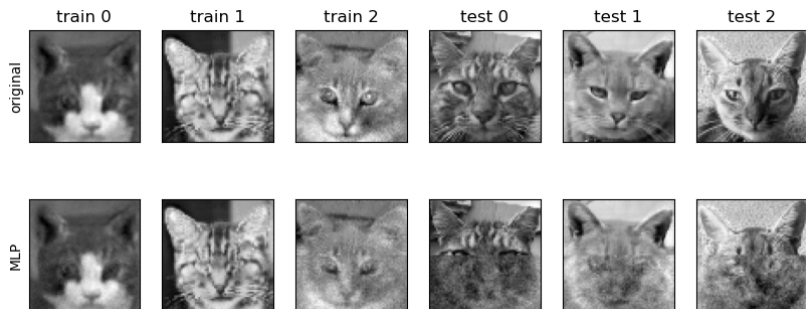


Figure: Predicting bottom half from top half, default settings.

- note that even the training set had issues doing to the prediction...
 - predictions can be quite cursed if eye happens to be in top half (e.g. grant us eyes)

MLPs: regression problem

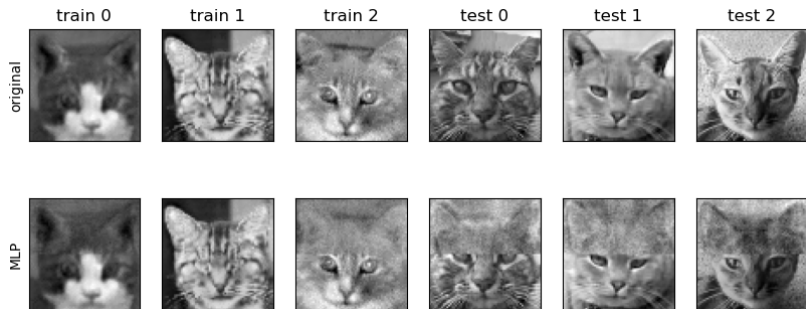


Figure: Predicting top half from bottom half, default settings.

- note that even the training set had issues doing to the prediction...
 - less cursed at least subjectively
 - at least somewhat continuous predictions?

MLPs: regression problem

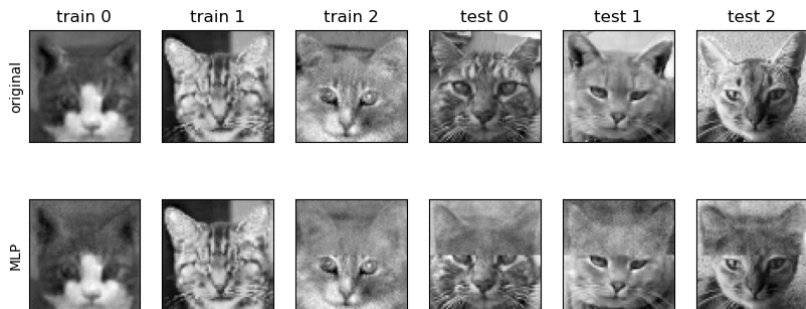


Figure: Predicting top half from bottom half, reasonably complicated.

- note that even the training set had issues doing to the prediction...
 - not doing much better with substantially increased complexity

Demonstration



Figure: Cursed beasts reconstruction (out of sample application of MLP).

- ▶ idea behind neural networks
→ ways to control over-fitting not considered here (e.g. **dropout**, **regularisation**)
- ▶ note: MLPs are fully-connected by default, dimensionality blows up quickly!
- ▶ need to cross-validate and tune hyper-parameters accordingly!

Moving to a Jupyter notebook →